each species had unalterable characteristics—continued to dominate Western thought.

## Great breakthroughs

Geological discoveries in the late 17th and early 18th centuries began to challenge the idea of essentialism. Geologists noted that some fossil species suddenly disappeared from the geological record, to be replaced by others, suggesting that organisms change over time, and even become extinct. In 1809, French naturalist Jean-Baptiste Lamarck proposed the first cohesive theory of evolution—the transmutation of species by the inheritance of acquired characteristics. However, some 50 years later it was Charles Darwin and Alfred Russel Wallace who developed the concept of evolution by means of natural selection, the theory that organisms evolve over the course of generations to adapt better to their environment. Darwin and Wallace did not understand the mechanism by which this happened, but Gregor Mendel's experiments on peas pointed to the role of hereditary factors later known as genes, representing another giant leap in evolutionary theory.

**"There are some 4 million different kinds of animals and plants in the world. Four million different solutions to the problems of staying alive."**

**David Attenborough**

## Making connections

Relationships between organisms and their environment, and between species, dominated early 20th-century ecological study. The concepts of food chains and food webs (who eats what in a particular habitat) and ecological niches (the role a species has in its environment) developed. In 1935, Arthur Tansley introduced the concept of the ecosystem—the interactive relationship between living organisms and the environment in